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Two Ordering Principles and the Area-Bound Theorem

Resonance, Entropy, and Their Geometric Resolution

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Abstract

In the physics of black holes, two ordering principles appear that seem to contradict each other: the **resonance principle** (ξ-recursion selects resonant configurations) and the **thermodynamic principle** (entropy seeks its maximum). This document shows that the contradiction is resolved through a separation of levels, and that at the evaporation endpoint the two principles are mutually enforcing. A geometric theorem (area-bound argument) is derived. The connection to Vopson's Second Law of Infodynamics is characterised as an honest translation. Three open proof gaps are explicitly identified.

1 The Two Ordering Principles

1.1 The resonance principle

In FFGFT, the ξ-recursion on T^4 selects the admissible winding configurations. Only configurations satisfying the recursion condition are stable — particles have discrete masses, not a continuum. The ξ-principle is order-creating: it generates structure, selects the resonant path, and defines the admissible state space before dynamics operate.

1.2 The thermodynamic principle

Thermodynamics seeks maximum entropy: disorder increases, structures equilibrate, everything flows toward the averaged state. This appears directly opposed to the resonance principle.

The apparent contradiction

The resonance principle favours order and structure. The thermodynamic principle favours equilibration and disorder. Both act simultaneously in nature. How is this possible?

2 The Area-Bound Theorem

2.1 The argument chain

The following argument is a **theorem**, not a postulate. It follows from three established quantities:

- **Holographic bound (Bekenstein):** N_max proportional A.

- **Hawking evaporation:** A proportional M^2 shrinks with mass.
- **EFGET lower bound:** $L_0 = xi \cdot L_P$ — state space ends at $N = 0$.

Area-bound theorem
From (1), (2), and (3) it follows necessarily: the **possible information capacity** of the horizon surface must decrease during evaporation. Not postulated — derived from the geometry.

$$N_{max}(M) \text{ proportional } M^2 \Rightarrow dN_{max}/dt < 0 \text{ (while the BH evaporates)}$$

2.2 Concentration as a consequence

Forced concentration
- The *possible* information (capacity N_max) decreases — geometrically forced.
- The *actual* topological charge Q is preserved — topological invariance.
- Therefore Q must be packed into ever fewer degrees of freedom: **concentration is enforced**.

At the remnant ($N = 0$): maximum charge concentration at minimum capacity. This is the resonance principle — not chosen, but inevitable.

No contradiction, but a causal chain
The entropic principle generates the conditions under which the resonance principle becomes inevitable. The one causes the other.

3 Python-Verified Key Numbers

3.1 Counter-directional behaviour: N_node vs. radiation entropy

Numerical verification (infodynamik.py): N_node proportional M^2 falls monotonically. Simultaneously S_rad rises with Page factor 4/3 — radiation carries away more entropy than the BH loses. For M = M_Sun:

$$k_B T_H / E_{max} \approx 5.8 \times 10^{-44},\ N_{node}(M_{Sun}) \approx 1.5 \times 10^{77},\ M_{crit} \approx 1.15 \times 10^{-13}\text{ kg}$$

3.2 Scaling comparison: capacity vs. naive charge count

M	N_max/Q	Status
1 M_Sun	6.9×10^{16}	consistent
10^{20} kg	3.5×10^{6}	consistent
10^{11} kg	3.5×10^{-3}	scissors cross

Meaning of the crossing
$Q = M/m_e$ exceeds N_max at $M \approx 2.9 \times 10^{13}$ kg — this is **not a physical contradiction** but the proof that M/m_e is the **wrong account**. It counts infallen particles, not topological invariants.

4 Connection to Vopson's Infodynamics

Vopson's Second Law: information entropy decreases or stays constant while thermodynamic entropy increases.

In FFGFT quantities: $N_{max}(M)$ proportional M^2 falls (capacity), S_{rad} rises — same directions as Vopson's law. **But the mechanism differs.**

Vopson	FFGFT
Information entropy decreases as a postulated law of nature	Capacity N_{max} decreases as a geometric theorem (area + Bekenstein + L_0)
Counts: information-bearing particle states	Counts: surface nodes proportional A proportional M^2

Honest translation

FFGFT reproduces the *phenomenology* of Vopson's law through a different mechanism. The two accounts count different things. The connection is a **translation**, not an identity.

This is a stronger epistemic position: FFGFT postulates nothing but derives the counter-directional behaviour from the geometry.

5 Three Open Proof Gaps

5.1 The correct topological charge

Open question: What is the conserved quantity Q in units comparable to N_{max}?

$Q = M/m_e$ is demonstrably wrong. Q must scale at most as M^2. Whether Q is identical to the Bekenstein entropy, sub-extensive, or something else — is the open question. Only then can $Q \leq N_{max}$ be verified as a theorem throughout evaporation.

5.2 The extremal principle

Open question: Can the resonance principle be formulated as a variational statement?

The Fibonacci convergence $\varphi^{-2n}/\sqrt{5} \approx \xi$ at $n \approx 8$ and the Lagrange stiffness term $-\frac{1}{2}m_T^2(\partial m)^2$ suggest an extremal principle. But the explicit Euler-Lagrange statement — a proof that the ξ fixed point is a minimum of an information functional $\square[\Psi]$ — does not yet exist in the corpus.

5.3 Definition of information entropy in FFGFT

Open question: How does FFGFT define information entropy precisely?

N_{node} currently serves as the information measure. But N_{node} is the holographic capacity (surface nodes), not the actually stored information. The clean separation of information capacity from information entropy is missing —

and without it a precise bridge to Vopson's law cannot be fully built.

Summary

Point	Statement
Apparent contradiction	Resonance principle (order) vs. thermodynamic principle (equilibration)
Resolution	Different levels: topological vs. thermodynamic
Theorem	Area bound + Bekenstein + L_0 => capacity necessarily decreases
Causal chain	Entropy drives evaporation => area shrinks => concentration enforced
Vopson bridge	Same phenomenology, different mechanism: translation, not identity
Account difference	Q = M/m_e (particles) ≠ N_max (surface) — two different counts
Open question 1	Correct topological charge Q: must Q ≤ N_max proportional M^2 hold
Open question 2	Extremal principle: ξ fixed point as minimum of an information functional
Open question 3	Clean definition of information entropy in FFGFT